

Pythagoras: see the connection

FREE PRACTICE SHEET / Show your working. Give units.

For a right-angled triangle: $a^2 + b^2 = c^2$.
 c is the hypotenuse, opposite the right angle.

- 1.** The shorter sides of a right-angled triangle are 6 cm and 8 cm.
Find the hypotenuse.

- 2.** The hypotenuse is 13 cm. One shorter side is 5 cm.
Find the other shorter side.

- 3.** A rectangle is 9 m long and 12 m wide.
Find the length of its diagonal.

- 4.** A ladder is 10 m long. Its base is 6 m from a vertical wall.
How far up the wall does it reach? Assume level ground.

- 5.** A triangle has sides of 7 cm, 8 cm and 11 cm.
Is it right-angled? Explain using a calculation.

Make every step make sense

ANSWERS & REASONING / Check the method as well as the result.

1. The hypotenuse is 10 cm.

$6^2 + 8^2 = 36 + 64 = 100$. The square root of 100 is 10.

2. The missing side is 12 cm.

$13^2 - 5^2 = 169 - 25 = 144$. The square root of 144 is 12.

3. The diagonal is 15 m.

$9^2 + 12^2 = 81 + 144 = 225$. The square root of 225 is 15.

4. The ladder reaches 8 m up the wall.

$10^2 - 6^2 = 100 - 36 = 64$. The square root of 64 is 8.

5. No, the triangle is not right-angled.

$7^2 + 8^2 = 113$, but $11^2 = 121$. They are not equal.

A useful check

The hypotenuse must be the longest side. If your answer makes a shorter side longer than the hypotenuse, revisit your calculation.